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Comparative 3D Depth Rendering System

Z-Buffer vs Painter's Algorithm for real-time 3D rendering

Computer Graphics Microproject | IT (AR/VR) | Harshith S

THE PROBLEM

A 3D scene is flattened onto a 2D screen, so many surfaces land on the same pixel. The renderer must pick the nearest one: hidden surface removal.

OBJECTIVE

Compare both algorithms in one live Python application: same scenes, same rasterizer, different visibility method. Measure speed, overdraw, memory and errors.

SYSTEM ARCHITECTURE

1 Scene: 3 test scenes built from triangles

2 Transform and cull: rotate, camera space, drop back faces

3 Project and shade: perspective divide, flat lighting

4 Shared NumPy rasterizer (same for both)

Z-Buffer path | **Painter's path**

5 Metrics and Pygame display

Z-BUFFER: test depth at every pixel

1 Clear the colour buffer; set the depth buffer to infinitely far

2 Rasterize each triangle into pixels

3 Compare each pixel's depth with the stored depth

4 If nearer, write colour and depth; otherwise discard

Correct for any triangle order, including intersections. Costs a depth buffer (300 KB at 320x240).

PAINTER'S: sort, then overwrite

1 Compute one depth per triangle (centroid or farthest vertex)

2 Sort all triangles far to near

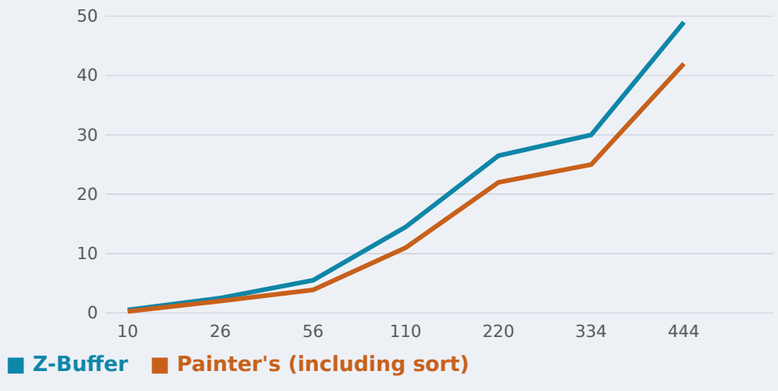
3 Paint in order; nearer triangles overwrite farther ones

No depth buffer needed. Fails on intersecting and cyclic overlaps, and paints hidden pixels.



Live output, scene 1 (intersecting cubes): Z-Buffer (left) vs Painter's (right). Painter's shows wrong pixels at the intersection.

RESULTS: FRAME TIME (ms) VS TRIANGLES DRAWN



6.1%
of pixels wrong in Painter's at 444 triangles

3.3x vs 1.8x
overdraw: Painter's vs Z-Buffer at 444 triangles

300 KB
depth buffer the Z-Buffer needs at 320x240

Conclusion: Z-Buffer is the robust general-purpose choice. Painter's is simpler and cheaper, but only reliable when objects never intersect or overlap in a cycle.